# Walker AI – Estimate Confidence Scoring Model Specification

The Estimate Confidence Scoring Model allows Walker AI to indicate how reliable an automatically generated renovation estimate is. Renovation estimates always involve uncertainty due to hidden conditions, structural variables, and incomplete user inputs. The confidence model measures these risks and adjusts the displayed confidence percentage accordingly.

## Purpose of the Confidence Score

The confidence score communicates estimate reliability to users and helps prevent false precision. It also flags projects that may require expert review or on■site inspection before pricing can be finalized.

## Confidence Score Range

| Confidence Score | Meaning |
|---|---|
| 85–95% | High confidence – minimal unknown factors, layout stable |
| 70–84% | Moderate confidence – some variables present |
| 50–69% | Low confidence – multiple unknowns or structural changes |
| Below 50% | Very low confidence – site inspection recommended |

## Primary Risk Factors Affecting Confidence

| Risk Factor | Confidence Impact |
|---|---|
| House built before 1980 | -10% |
| Structural wall modification | -15% |
| Drain relocation | -12% |
| Gas line change | -10% |
| Exterior penetration (vent/window/door) | -8% |
| Basement project | -6% |
| Three or more 'Not sure' answers | -10% |

## Confidence Score Calculation Example

Base confidence begins at 90%. Risk factors subtract from the score depending on project conditions.

| Condition | Impact |
|---|---|
| Base confidence | 90% |

| | |
|---|---|
| Drain relocation | -12% |
| Basement renovation | -6% |
| Two 'Not sure' answers | -4% |
| Final confidence | 68% |

## Developer Implementation Logic (Example)

Example pseudocode implementation:

```
confidence = 90

if house_age < 1980:
confidence -= 10

if structural_changes:
confidence -= 15

if drain_relocation:
confidence -= 12

if exterior_penetration:
confidence -= 8

if basement_project:
confidence -= 6

if unknown_answers >= 3:
confidence -= 10
```

## User Display Example

Estimated Renovation Cost: $21,000 – $24,500
Estimate Confidence: 68%

A message can be displayed recommending expert review when the confidence score drops below a defined threshold.